

Notes: Tetlock (RFS, 2010)

Does Public Financial News Resolve Asymmetric Information?

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1 Big picture

- **Question.** Does public financial news merely attract investor attention, or does it genuinely *resolve asymmetric information* across traders?
- **Setting.** The paper studies firm-specific Dow Jones and *Wall Street Journal* news from 1979 to 2007 and links those stories to daily returns, turnover, analyst data, institutional ownership, and intraday trade data.
- **Core challenge.** News can coincide with information arrival, liquidity shocks, disagreement, and investor attention. A reduced-form news-return correlation alone cannot tell us which force is dominant.
- **Main theoretical idea.** Before the public news release, informed investors hold both a private signal about fundamentals and a persistent liquidity shock. When the news arrives, uninformed traders learn the signal and infer more about the liquidity shock. This changes both pricing and trading behavior around the event.
- **Key theoretical predictions.**
 - News-event returns should positively predict postnews returns.
 - High-volume news should predict even more postnews momentum.
 - The correlation between absolute returns and trading volume should rise around news.
 - The price impact of signed order flow should fall around news.
- **Main empirical strategy.** Estimate daily cross-sectional return-predictability regressions comparing news and nonnews days, then use event-time tests around the same news events to track the volume–return correlation and intraday price impact before and after public news.
- **Main empirical message.** The data support the asymmetric-information interpretation. News days exhibit substantially smaller short-run reversals, stronger volume-induced momentum, stronger contemporaneous return–volume comovement, and lower price impact.
- **Cross-sectional message.** The results are generally strongest for **small, illiquid, and high-message-count** news events, exactly where one would expect information asymmetry to matter most.
- **Why this matters.** The paper links textual news data to classic market microstructure questions. It shows that public news is not just an attention shock; it changes who knows what, and therefore changes the way liquidity is supplied.
- **Economic takeaway.** Public news helps previously uninformed traders absorb outstanding liquidity shocks. That is why prices continue in the same direction after the news, why trading volume and price moves become more tightly linked, and why the price impact of order flow temporarily falls.

2 Data and measurement (key objects)

2.1 Primary news data and sample construction

- **News source.** The primary data source is the Dow Jones archive, which contains all DJ News Service and *Wall Street Journal* stories from 1979 to 2007.

- **Firm-day news measure.** Define

$$News_{it} = 1$$

if firm i appears in any Dow Jones or *WSJ* story between the close of day $t - 1$ and the close of day t .

- **Sample restrictions.** The paper focuses on U.S. common shares listed on NYSE, NASDAQ, or Amex, with prices above \$5 and positive trading volume on all days from $t - 60$ to $t - 1$.
- **Matched data.** News and returns are matched to CRSP, accounting data from Compustat, analyst forecast data from I/B/E/S, institutional holdings from Thomson 13F, and transaction data from ISSM/TAQ.
- **Coverage.** After restrictions, the sample contains 13,842 unique firms, of which 9,452 appear in the news at least once.

Interpretation. The paper is not about a handful of heavily covered firms only. It uses a broad cross-section, but it still imposes liquidity filters so the return and volume measures are meaningful.

2.2 Coverage over time

- News coverage rises strongly over time.
- In the 1980s, roughly 2%–5% of eligible firms appear in the archive on a typical day.
- In the post-2000 period, the daily coverage rate is roughly 20%–35%.
- For the firm at the 90th percentile of media coverage, news appears on roughly 10% of trading days in 1980 and roughly 60% of trading days in 2007.

Interpretation. Figure 1 matters because it shows that media coverage changes a lot over time, so the paper must guard carefully against subperiod effects and data-quality changes.

2.3 Main return and volume measures

- **Day- t return.**

$$Ret_{it} = r_{it} - r_t^m,$$

where the benchmark is the value-weighted market return.

- **Postnews return.** The main dependent variable is the raw return from day $t + 2$ through day $t + 10$:

$$Ret_{i,t+2,t+10}.$$

Day $t + 1$ is omitted to reduce bid–ask bounce.

- **Abnormal turnover.** The volume variable is the Campbell–Grossman–Wang style abnormal turnover measure:

$$Turn_{it} = \log(\text{turnover}_{it}) - \frac{1}{60} \sum_{j=1}^{60} \log(\text{turnover}_{i,t-j}).$$

- **Firm controls.** The main control set includes size, book-to-market, annual momentum excluding the most recent month, and recent return volatility.

Interpretation. The paper is designed to isolate short-horizon expected-return patterns after public news, not to estimate unconditional factor premia.

2.4 News informativeness proxies

- **Earnings-news indicator.** The paper flags stories mentioning “earnings” or the stem “earn.”
- **Message count.** Let Msg_{it} denote the log number of distinct newswire messages for firm i on day t .

Interpretation. The number of wire messages is used as a proxy for the informativeness and importance of the event. A story broken into many wire messages is more likely to be timely, important, or detailed.

2.5 Firm characteristics used in heterogeneity tests

The paper sorts firms by:

- size,
- Amihud illiquidity,
- analyst coverage,
- analyst forecast dispersion,
- institutional ownership.

Interpretation. These sorts help distinguish the information-based mechanism from alternatives based on investor attention or differences of opinion.

2.6 Intraday price-impact measure

For each firm-day, signed order flow and returns are aggregated into 5-minute intervals. Price impact is then measured as

$$PI_{it} = \frac{\text{Cov}(r_{i\tau}, OF_{i\tau})}{\text{Var}(OF_{i\tau})},$$

where τ indexes 5-minute intervals and signed order flow is constructed using the Lee–Ready algorithm.

Interpretation. This is an empirical proxy for the price impact of informed trading. If public news resolves asymmetric information, the same signed order flow should move prices less on the news day.

2.7 Measurement issues and how the paper handles them

- **Firm-code measurement error before 1997.** The author notes that DJ firm codes before November 1996 are imperfect and may be back-filled, creating possible survivorship bias.
- **Response.** The paper reports subperiod results and argues that the main findings hold both before and after 1996 and are often stronger after 1996.
- **Microstructure bias.** Day $t + 1$ is omitted from the postnews return window.
- **Composition effects.** For the volume-correlation and price-impact exercises, the paper studies the same news firms in event time rather than relying only on unconditional news-versus-no-news comparisons.

3 Models and empirical specifications (all equations + interpretation)

3.1 Stylized model: environment and timing

The model has three trading periods, two investor groups, one risky asset, and one risk-free asset.

- Group i is temporarily informed and also receives a privately observed liquidity shock.
- Group u is relatively uninformed but rational.
- Investors have CARA utility and behave myopically.

The liquidating dividend is

$$d_3 = d + s_1 + e_1 + e_2 + e_3,$$

where $s_1 \sim N(0, V_s)$ is the private signal observed by informed traders in period 1 and each $e_t \sim N(0, V_e)$ is publicly revealed in period t .

Informed investors also receive a persistent liquidity shock

$$n_1 \sim N(0, V_n),$$

which survives until after period 2. In period 2, a public news announcement reveals the signal s_1 to uninformed investors.

Interpretation. News arrives after informed traders have already used both information and liquidity needs to trade. That is what makes the event informative about both expected cash flows and order-flow motives.

3.2 Demand and equilibrium prices

With CARA-normal demand, informed demand per investor is

$$x_{it} = \frac{E_{it}(p_{t+1}) - p_t}{\text{Var}_{it}(p_{t+1})}. \quad (1)$$

Market clearing in period 2 implies

$$p_2 = d - V_e + e_1 + e_2 + s_1 + mV_en_1. \quad (2)$$

In period 1 the equilibrium price is linear:

$$p_1 = a + e_1 + b_s s_1 + b_n n_1, \quad (3)$$

with coefficients satisfying

$$0 < b_s < 1, \quad mV_e < b_n < 2mV_e.$$

The initial price is

$$p_0 = d - 3V_e - (b_s^2 V_s + b_n^2 V_n). \quad (4)$$

Interpretation. The key feature is that the period-1 price partially reflects the private signal and partially reflects the hidden liquidity shock. Once news reveals s_1 , uninformed traders can infer much more about n_1 as well.

3.3 Abnormal returns and postnews predictability

The period-2 and period-3 abnormal returns are

$$r_2 = (1 - b_s)s_1 + (mV_e - b_n)n_1 + e_2, \quad (5)$$

$$r_3 = e_3 - mV_en_1. \quad (6)$$

Hence the return-predictability coefficient is

$$\frac{\text{Cov}(r_3, r_2)}{\text{Var}(r_2)} = \frac{mV_e(b_n - mV_e)V_n}{(1 - b_s)V_s + V_e + (b_n - mV_e)^2 V_n} > 0. \quad (7)$$

Proposition 1. The regression coefficient of postnews returns on news-event returns increases with the informativeness of news.

Interpretation. News allows uninformed investors to absorb part of the old liquidity shock in period 2; the remainder dissipates in period 3. That generates positive return autocorrelation after news.

3.4 Trading volume on the news day

Abnormal trading volume in the news period is

$$T_2 = |m(x_2^i - x_1^i - E(x_2^i - x_1^i))| \quad (8)$$

$$= \frac{m}{V_e} |[1 - b_s]s_1 + [2mV_e - b_n]n_1|. \quad (9)$$

Proposition 2. Expected trading volume increases with the informativeness of news.

Interpretation. High news-day volume is informative because more informative public news induces more accommodation of the liquidity shock.

3.5 Correlation between absolute returns and volume

The model implies that the correlation between volume and absolute returns rises with informativeness:

$$\text{Corr}(T_2, |r_2|) = \frac{2 \sin(\arccos(-\rho)) + 2\rho \arccos(-\rho) - \rho\pi - 2}{\pi - 2}, \quad (10)$$

where $\rho = \text{Corr}(m(x_2^i - x_1^i - E(x_2^i - x_1^i)), r_2)$.

Proposition 3. The correlation between trading volume and absolute returns in the news-event period increases with news informativeness.

Interpretation. More informative news both moves expected returns more and induces more accommodation of the liquidity shock, so large returns and high trading volume arrive together.

3.6 Price impact

The model's final theoretical prediction is about price impact:

$$\frac{\text{Cov}(r_2, x_2^i - x_1^i - E(x_2^i - x_1^i))}{\text{Var}(x_2^i - x_1^i - E(x_2^i - x_1^i))} = \left(\frac{V_e}{m} \right) \frac{-(1 - b_s)^2 V_s + (2mV_e - b_n)(b_n - mV_e)V_n}{(1 - b_s)^2 V_s + (2mV_e - b_n)^2 V_n}. \quad (11)$$

Proposition 4. Price impact in the news period decreases with news informativeness.

Interpretation. Before news is released, informed traders can profit from superior information. Once the news arrives, that informational edge shrinks, so their order flow has less price impact.

3.7 Main return-reversal regression

The main empirical specification is

$$Ret_{i,t+2,t+10} = a + b_1 Ret_{it} + b_2 news_{it} Ret_{it} + b_3 Turn_{it} Ret_{it} \quad (12)$$

$$+ b_4 news_{it} Turn_{it} Ret_{it} + c Controls_{it} + e_{it}. \quad (13)$$

Interpretation.

- $b_1 < 0$ measures unconditional short-run return reversal.
- $b_2 > 0$ means news makes that reversal smaller.
- $b_3 > 0$ means high abnormal turnover predicts weaker reversal or even momentum.
- $b_4 > 0$ means turnover is especially informative when it occurs on a news day.

3.8 Regression with explicit news-informativeness proxy

The paper augments the baseline with the number of newswire messages:

$$Ret_{i,t+2,t+10} = a + b_1 Ret_{it} + b_2 news_{it} Ret_{it} + b_3 Turn_{it} Ret_{it} \quad (14)$$

$$+ b_4 news_{it} Turn_{it} Ret_{it} + d_1 msgSA_{it} Ret_{it} + d_2 msgSA_{it} + c Controls_{it} + e_{it}. \quad (15)$$

Interpretation. If message count is a better proxy for information content than raw turnover, then $msgSA_{it} Ret_{it}$ should subsume some of the predictive power of $news_{it} Turn_{it} Ret_{it}$.

3.9 Volume–absolute return regression

To study contemporaneous comovement, the paper estimates

$$Turn_{it} = Intercept + Slope \cdot |Ret_{it}| + e_{it}. \quad (16)$$

This is estimated on news days, nonnews days, and in event time for firms that experienced news exactly $k \in \{-10, \dots, 10\}$ days earlier.

Interpretation. The event-time version asks whether the *same firms* have unusually strong return–volume comovement exactly when public news arrives.

3.10 Price-impact event study

Using 5-minute data, the paper computes average price impact on day $t + k$ for firms with a news event on day t and traces the path for $k = -10$ to $+10$.

Interpretation. If public news resolves asymmetric information, price impact should dip on the news day and rebound afterward.

4 Identification and instruments (what makes the estimates credible)

4.1 Where identification comes from

This is not an IV paper. Identification comes from a combination of:

- comparing news and nonnews days,
- daily cross-sectional regressions with rich controls,
- event-time comparisons around the same news events,
- and heterogeneity by firm and news characteristics.

Interpretation. The central question is not “what is the average effect of a news dummy?” but rather whether the full bundle of news-day predictions lines up with the asymmetric-information mechanism.

4.2 Why the return tests are convincing

- The dependent variable starts at $t + 2$, which weakens the concern that results are just bid–ask bounce.
- Controls absorb several standard return-predictability effects.
- The regressions are estimated daily in Fama–MacBeth style, and standard errors are Newey–West adjusted.

Interpretation. The paper is explicitly designed to show that short-run news predictability is not merely a by-product of known reversal or momentum patterns.

4.3 Why event time matters

The event-study design for Tables 5–7 and Figures 3–4 is essential.

- News firms differ from nonnews firms in size, liquidity, and analyst coverage.
- Looking at the same firms before, during, and after the news event reduces the concern that unconditional news/nonnews comparisons are driven by composition differences.

4.4 Why message count is useful

The variable Msg_{it} is not perfect, but it is valuable because it measures informativeness more directly than turnover.

Interpretation. If turnover only proxies for information arrival, then once a better news-content proxy is included, the extra predictive power of turnover should shrink. That is exactly what the paper finds.

4.5 How the paper discriminates against alternative explanations

- **Pure liquidity-shock story.** If news mostly signaled liquidity shocks, news should predict stronger reversal, not weaker reversal.
- **Attention story.** Attention can explain some rise in turnover, but it does not naturally explain lower price impact and the full set of return-predictability patterns.
- **Differences-of-opinion story.** If disagreement were central, one might expect stronger effects where forecast dispersion is high. The evidence is not especially supportive.

4.6 Why small and illiquid stocks are informative

The paper repeatedly finds stronger news effects among small and illiquid stocks.

Interpretation. This strengthens the asymmetric-information view because those are exactly the firms where private information problems should be most severe.

4.7 Main limitations

- News is measured imperfectly, especially before late 1996.
- The model is stylized and is not a full dynamic microstructure model of all intraday mechanisms.
- Signed order flow is only a noisy proxy for informed trading.
- Some behavioral attention effects may still operate in the data even if they do not seem to be the primary force.

4.8 Relation to the literature

The paper connects four literatures:

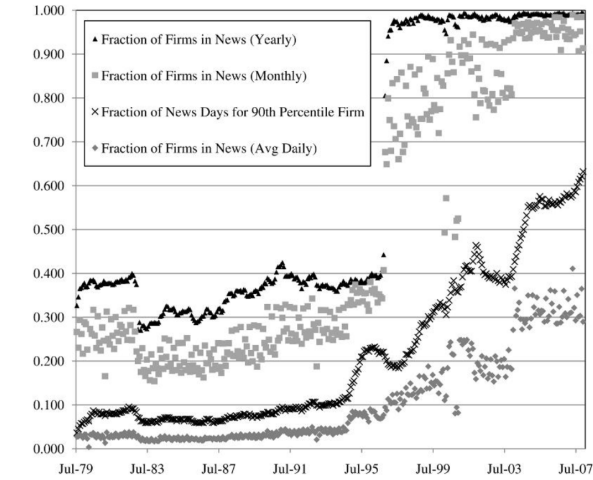


Figure 1

Media coverage across firms and over time

Figure 1 depicts how four media coverage measures change from 1979 to 2007. The bottom line is the monthly average of the daily fraction of sample eligible firms that have news in the DJ archive—see text for sample construction. The other three lines are three long-horizon coverage measures that include media coverage on trading days that meet the sample inclusion criteria. The top two lines represent the fraction of firms with at least one news story in the current month, and the fraction of firms with news in the most recent 12 months. The second line from the bottom is the fraction of trading days in the most recent 12 months that a firm appears in the news for the firm at the 90th percentile. This line represents news coverage for the most widely followed firms.

- short-run return reversals and post-event drift,
- news and expected returns,
- microstructure models with informed and liquidity traders,
- contemporaneous relations between returns, volume, and price impact.

Its distinctive contribution is to tie all four together using the idea that public news resolves asymmetric information.

5 Tables and figures

5.1 Figure 1: Media coverage over time

Read:

- Figure 1 shows that the fraction of eligible firms appearing in the news rises dramatically from the early 1980s to the 2000s.
- This matters because any empirical design using news has to be robust to large changes in coverage intensity over time.
- The paper's pre/post-1997 robustness checks are therefore not cosmetic; they are essential.

5.2 Table 1 and Figure 2: News and return reversals

Read:

- The baseline reversal coefficient on Ret_{it} is strongly negative, around -0.10 .

Table 1
The impact of news on return reversal and volume-induced return momentum

Independent Variable	All Days	Excluding Nonearnings News	Excluding Earnings News	1979–1996	1997–2007
Ret_{it}	-0.098** (0.003)	-0.084** (0.004)	-0.100** (0.003)	-0.125** (0.003)	-0.067** (0.005)
$news_{it} * Ret_{it}$	0.042** (0.004)	0.084** (0.007)	0.029** (0.004)	0.038** (0.007)	0.043** (0.004)
$Turn_{it} * Ret_{it}$	0.0088** (0.0012)	0.0058** (0.0015)	0.0084** (0.0012)	0.0120** (0.0013)	0.0019 (0.0024)
$news_{it} * Turn_{it} * Ret_{it}$	0.0188** (0.0030)	0.0091* (0.0043)	0.0174** (0.0035)	0.0172** (0.0049)	0.0193** (0.0036)
$size_{it} * Ret_{it}$	-0.0043** (0.0010)	-0.0005 (0.0012)	-0.0051** (0.0010)	-0.0092** (0.0012)	-0.0025 (0.0018)
$News_{it}$	-0.0001 (0.0001)	-0.0005* (0.0003)	0.0000 (0.0001)	0.0002 (0.0002)	-0.0001 (0.0001)
$news_{it} * Turn_{it}$	-0.0007** (0.0001)	-0.0005* (0.0002)	-0.0003* (0.0001)	-0.0007** (0.0002)	-0.0007** (0.0002)
$Turn_{it}$	0.0023** (0.0001)	0.0020** (0.0002)	0.0023** (0.0001)	0.0022** (0.0001)	0.0024** (0.0001)
$Size_{it}$	-0.0003** (0.0001)	-0.0003 (0.0001)	-0.0003** (0.0001)	-0.0003 (0.0001)	-0.0004* (0.0002)
BM_{it}	0.0018** (0.0004)	0.0020** (0.0005)	0.0019** (0.0004)	0.0019** (0.0006)	0.0018** (0.0006)
Mom_{it}	0.0066** (0.0007)	0.0060** (0.0008)	0.0067** (0.0007)	0.0078** (0.0009)	0.0055** (0.0011)
$TVol_{it}$	-0.115** (0.023)	-0.099** (0.029)	-0.115** (0.023)	-0.132** (0.023)	-0.101** (0.038)
Average R^2	0.047	0.045	0.047	0.047	0.047
Avg obs per day	2279	2271	2222	1900	2872
Days	7098	4432	7096	4335	2763

Table 1 reports all of the estimated coefficients from the regression in Equation (15) in the text:

$$Ret_{i,t+2,t+10} = a + b_1 * Ret_{it} + b_2 * news_{it} * Ret_{it} + b_3 * Turn_{it} * Ret_{it} + b_4 * news_{it} * Turn_{it} * Ret_{it} + c * Controls_{it} + e_{it} \text{ for all } i \text{ on day } t,$$

where $Controls_{it} = [News_{it}, size_{it} * Ret_{it}, news_{it} * Turn_{it}, Turn_{it}, Size_{it}, BM_{it}, Mom_{it}, TVol_{it}]^T$. The dependent variable ($Ret_{i,t+2,t+10}$) is the firm's raw return from day $t + 2$ to $t + 10$. The independent variables include day- t market-adjusted returns (Ret_{it}); the news indicator ($News_{it}$); abnormal turnover ($Turn_{it}$); interactions between news and returns ($news_{it} * Ret_{it}$); turnover and returns ($Turn_{it} * Ret_{it}$); news, turnover, and returns ($news_{it} * Turn_{it} * Ret_{it}$); firm size and returns ($size_{it} * Ret_{it}$); and news and turnover ($news_{it} * Turn_{it}$). Other controls include size ($Size_{it}$), book-to-market (BM_{it}), annual return momentum (Mom_{it}), and monthly return volatility ($TVol_{it}$). I estimate Equation (15) daily using all firms much like Fama and MacBeth (1973). The point estimate is the time series average of the daily regression coefficients. The Newey and West (1987) standard errors in parentheses below are robust to autocorrelation up to 10 daily lags and heteroscedasticity in the daily coefficient estimates. The five columns denote subsamples used to estimate Equation (15): all trading days, excluding nonearnings news, excluding earnings news, and pre- and post-1997. The * and ** symbols denote statistical significance at the 5% and 1% levels.

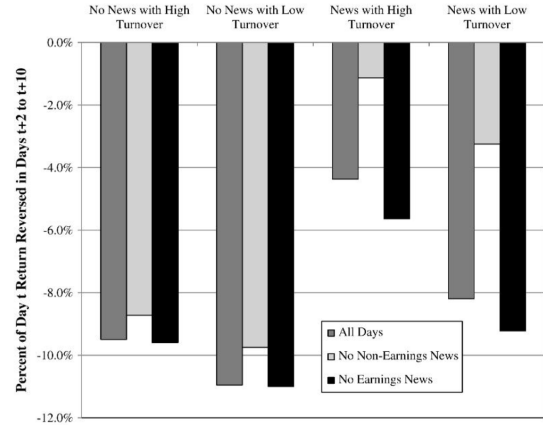


Figure 2
The impact of news on return reversal and volume-induced momentum

Figure 2 shows how the percentage reversal of a daily return depends on the news and trading volume observed on that day. The calculations in the figure come directly from the regression estimates of the four coefficients (b_1 , b_2 , b_3 , and b_4) in Equation (15):

$$Ret_{i,t+2,t+10} = a + b_1 * Ret_{it} + b_2 * news_{it} * Ret_{it} + b_3 * Turn_{it} * Ret_{it} + b_4 * news_{it} * Turn_{it} * Ret_{it} + c * Controls_{it} + e_{it} \text{ for all } i, t.$$

Table 1 reports these four coefficients on the variables Ret_{it} , $news_{it} * Ret_{it}$, $Turn_{it} * Ret_{it}$, and $news_{it} * Turn_{it} * Ret_{it}$. I compute the predicted value of the dependent variable when there is no news with high turnover, no news with low turnover, news with high turnover, and news with low turnover in the four sets of bars below. The value of $news_{it}$ is equal to either 0.702 or -0.099, depending on news or no news. For high or low turnover, the value of abnormal turnover is the 90th or 10th percentile of the turnover distribution conditional on news. The dark gray, light gray, and black bars denote three sets of coefficient estimates for the news- and volume-related return momentum in three subsamples: all trading days, excluding earnings news, and excluding nonearnings news. Columns 1, 2, and 3 in Table 1 report these coefficient values.

- The coefficient on $news_{it} Ret_{it}$ is strongly positive, around 0.04, showing that reversals are much smaller on news days.
- The coefficient on $news_{it} Turn_{it} Ret_{it}$ is also positive, showing that high-volume news predicts even more momentum.
- Figure 2 visualizes the same idea: high-volume news days exhibit the weakest reversal and in some cases clear short-run continuation.

Economic message. News makes returns more informative, and high-volume news makes them more informative still.

5.3 Table 3: Heterogeneity by firm and news characteristics

Read:

- The reduction in reversal from news is generally stronger for small and illiquid firms.
- The interaction between news, turnover, and returns is especially strong in illiquid and small firms.
- Once the sample is split by message count, much of the extra role of turnover disappears, indicating that turnover partly proxies for news informativeness.

Economic message. The results are strongest exactly where asymmetric information should matter most.

Sort Variable	High Sort Variable		Low Sort Variable		High-Low Sort Variable	
	<i>newsRet</i>	<i>newsTurnRet</i>	<i>newsRet</i>	<i>newsTurnRet</i>	<i>newsRet</i>	<i>newsTurnRet</i>
<i>Size_{it}</i>	0.024** (0.008)	0.025** (0.005)	0.024* (0.011)	0.048** (0.008)	-0.005** (0.002)	0.019** (0.006)
<i>IlliquiditySA_{it}</i>	0.038** (0.008)	-0.014** (0.002)	0.042** (0.007)	0.036** (0.010)	0.013 (0.008)	-0.002** (0.005)
<i>AnalystSA_{it}</i>	0.036** (0.008)	0.010** (0.003)	0.024** (0.007)	0.042** (0.010)	0.000 (0.003)	0.035** (0.004)
<i>DispersSA_{it}</i>	0.023* (0.010)	0.007 (0.004)	0.004 (0.010)	0.024** (0.008)	0.019** (0.004)	-0.031** (0.012)
<i>InstOwnSA_{it}</i>	0.043** (0.007)	0.010** (0.002)	0.017** (0.006)	0.043** (0.011)	-0.005 (0.004)	0.022* (0.015)
<i>MsgSA_{it}</i>	0.095** (0.008)	0.003 (0.002)	0.004 (0.004)	0.019** (0.005)	0.008** (0.001)	0.079** (0.001)

Table 3 reports the estimated coefficients (b_2, b_3 , and b_4) on $news_{it} * Ret_{it}$, $Turn_{it} * Ret_{it}$, and $news_{it} * Turn_{it} * Ret_{it}$ from the regression in Equation (15): $Ret_{i,t+2,t+10} = a + b_1 * Ret_{it} + b_2 * news_{it} * Ret_{it} + b_3 * Turn_{it} * Ret_{it} + b_4 * news_{it} * Turn_{it} * Ret_{it} + c + Controls_{it} + e_{it}$ for all i , where $Controls_{it} = [News_{it}, size_{it} * Ret_{it}, news_{it} * Turn_{it}, Turn_{it} * Size_{it}, BM_{it}, Mom_{it}, TVol_{it}]^T$. The dependent variable ($Ret_{i,t+2,t+10}$) is the firm's raw return from day $t + 2$ to $t + 10$. The independent variables include day- t market-adjusted returns (Ret_{it}), the news indicator ($News_{it}$), daily abnormal turnover ($Turn_{it}$), interactions between news and returns ($news_{it} * Ret_{it}$), turnover and returns ($Turn_{it} * Ret_{it}$), news, turnover, and returns ($news_{it} * Turn_{it} * Ret_{it}$), firm size and returns ($size_{it} * Ret_{it}$), and news and turnover ($news_{it} * Turn_{it}$). Other controls include size ($Size_{it}$), book-to-market (BM_{it}), annual return momentum (Mom_{it}), and monthly return volatility ($TVol_{it}$). I estimate Equation (15) daily using all firms much like Fama and MacBeth (1973). The full sample coefficient estimate is the time series average of the daily cross-sectional regression coefficients. The Newey and West (1987) standard errors in parentheses below are robust to autocorrelation up to 10 daily lags and heteroscedasticity in the daily coefficient estimates. The six rows denote six characteristics used to form subsamples in which I estimate Equation (15). The characteristics are firm size ($Size_{it}$), and five size-adjusted characteristics: illiquidity ($IlliquiditySA_{it}$), analyst coverage ($AnalystSA_{it}$), analyst forecast dispersion ($DispersSA_{it}$), institutional ownership ($InstOwnSA_{it}$), and newswire messages ($MsgSA_{it}$). The columns show the estimated $news_{it} * Ret_{it}$, $Turn_{it} * Ret_{it}$, and $news_{it} * Turn_{it} * Ret_{it}$ coefficients in the top characteristic-sorted quintile subsample, the bottom quintile subsample, and the difference in the coefficients between the top and bottom quintiles. The * and ** symbols denote statistical significance at the 5% and 1% levels.

Independent Variable	All Days	1979–1996	1997–2007
<i>Ret_{it}</i>	-0.098** (0.003)	-0.124** (0.003)	-0.068** (0.005)
<i>news_{it} * Ret_{it}</i>	0.042** (0.004)	0.035** (0.008)	0.043** (0.004)
<i>msgSA_{it} * Ret_{it}</i>	0.028** (0.004)	0.075** (0.013)	0.027** (0.004)
<i>Turn_{it} * Ret_{it}</i>	0.0081** (0.0012)	0.0117** (0.0013)	0.0000 (0.0025)
<i>news_{it} * Turn_{it} * Ret_{it}</i>	0.0100** (0.0032)	0.0105* (0.0054)	0.0099** (0.0038)
<i>News_{it}</i>	-0.0001 (0.0001)	0.0002 (0.0002)	-0.0002 (0.0001)
<i>msgSA_{it}</i>	-0.0007** (0.0001)	-0.0006 (0.0004)	-0.0007** (0.0001)
Average R^2	0.048	0.048	0.048
Avg obs per day	2280	1901	2873
Days	7082	4319	2763

Table 4 reports the estimated coefficients from the regression specification in Equation (16) in the text:

$$Ret_{i,t+2,t+10} = a + b_1 * Ret_{it} + b_2 * news_{it} * Ret_{it} + b_3 * Turn_{it} * Ret_{it} + b_4 * news_{it} * Turn_{it} * Ret_{it} + d_1 * msgSA_{it} * Ret_{it} + d_2 * msgSA_{it} + c * Controls_{it} + e_{it} \text{ for all } i \text{ on each day } t,$$

where $Controls_{it} = [News_{it}, size_{it} * Ret_{it}, news_{it} * Turn_{it}, Turn_{it} * Size_{it}, BM_{it}, Mom_{it}, TVol_{it}]^T$.

The return reversal, news-related reversal, message-related reversal, and word-length-related reversal coefficients (b_1, b_2, d_1 , and d_2) are the primary focus. The dependent variable ($Ret_{i,t+2,t+10}$) is the firm's raw return from day $t + 2$ to $t + 10$. The independent variables include day- t market-adjusted returns (Ret_{it}), the news indicator ($News_{it}$), interactions between news and returns ($news_{it} * Ret_{it}$), firm size and returns ($size_{it} * Ret_{it}$), turnover and returns ($Turn_{it} * Ret_{it}$), and news and turnover ($news_{it} * Turn_{it}$). The $msgSA_{it}$ variable is the log of the number of newswire messages for firm i on day t , adjusted for the average value for each size quintile. Other controls not shown include size ($Size_{it}$), book-to-market (BM_{it}), annual return momentum (Mom_{it}), monthly return volatility ($TVol_{it}$), and daily abnormal turnover ($Turn_{it}$). Lowercase variables denote demeaned variables. I estimate Equation (16) daily using all firms much like Fama and MacBeth (1973). The coefficient estimate is the time series average of the daily cross-sectional regression coefficients. The Newey and West (1987) standard errors in parentheses below are robust to autocorrelation up to 10 daily lags and heteroscedasticity in the daily coefficient estimates. The three columns denote estimates of Equation (16) in three subsamples: all trading days, pre-1997, and post-1997. The * and ** symbols denote statistical significance at the 5% and 1% levels.

5.4 Table 4: Message count and return predictability

Read:

- Adding $msgSA_{it}Ret_{it}$ leaves the baseline news effect intact but reduces the magnitude of $news_{it}Turn_{it}Ret_{it}$.
- The message-count interaction is positive and highly significant.
- A large increase in message count produces an economically meaningful reduction in return reversal.

Economic message. The more informative the public signal, the more the event looks like information resolution rather than a simple liquidity event.

5.5 Table 5 and Figure 3: Volume–absolute return correlation

Read:

- The slope of turnover on $|Ret|$ is much larger on news days than on nonnews days.
- The regression R^2 rises sharply on news days and is especially high for earnings news.
- Figure 3 shows that the correlation peaks at the news event and falls both before and after the event.

Economic message. News events temporarily strengthen the link between price discovery and trading activity.

5.6 Table 6: Which news events generate the strongest correlation effects?

Read:

Table 5
News and the contemporaneous volume-absolute return correlation

Independent Variable	All	News	No News	Nonearnings News	Earnings News
$ Ret_{it} $	11.9** (0.07)	14.4** (0.11)	11.3** (0.08)	13.0** (0.12)	13.9** (0.14)
Constant	-0.205** (0.004)	-0.068** (0.004)	-0.215** (0.004)	-0.080** (0.004)	0.078** (0.005)
Average R^2	0.079	0.212	0.060	0.172	0.366
Avg obs per day	2532	315	2217	381	93
Days	7132	7132	7132	4434	4434

Table 5 reports regression coefficients and correlation coefficients based on Equation (17) in the text:

$$Turn_{it} = Intercept + Slope * |Ret_{it}| + e_{it} \text{ for all } i \text{ on each day } t.$$

The dependent variable is abnormal daily turnover ($Turn_{it}$); and the independent variable is absolute daily returns $|Ret_{it}|$. To control for variables aside from information asymmetry that could affect the correlation between absolute returns and volume, I examine *changes* in the correlation before and after the same news events in Panel B. I estimate Equation (17) using daily cross-sectional regressions. In parentheses, I report Newey and West (1987) standard errors that allow for autocorrelation of up to three daily lags in the time series of coefficients. Panel A in Table 5 reports the average slope, intercept, R^2 , and correlation coefficients for these monthly regressions. The five columns denote five different subsamples in which I estimate Equation (17): all days, days with news, days without news, days with nonearnings-related news, and days with earnings-related news. The * and ** symbols denote statistical significance at the 5% and 1% levels. To avoid clutter, I suppress all asterisks for the raw correlations.

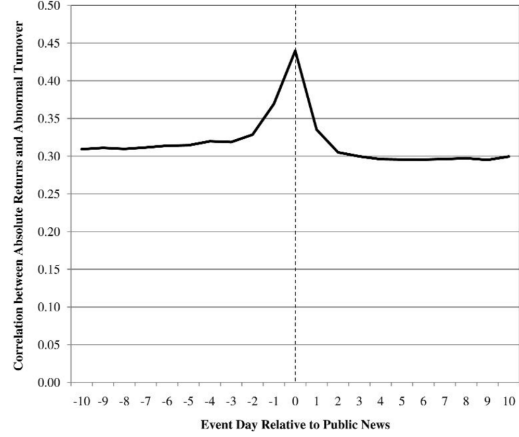


Figure 3
The correlation between absolute returns and volume around public news

Figure 3 reports the correlation between absolute returns and abnormal turnover on 21 different days, ranging from 10 days before a news event occurs to 10 days after the event. The correlation coefficients come from daily cross-sectional regressions based on Equation (17):

$$Turn_{it} = Intercept + Slope * |Ret_{it}| + e_{it} \text{ for all } i \text{ on day } t.$$

The independent variable is absolute daily returns $|Ret_{it}|$; and the dependent variable is abnormal daily turnover ($Turn_{it}$). To estimate the correlation between absolute returns and volume k days after a news event, I estimate Equation (17) using only observations in which a firm experienced a news event k days ago. Repeating this procedure for integer values of k ranging from -10 to +10 generates the figure below. See Table 5 for further estimation details.

Table 6
Volume-absolute return correlation sorted by firm and news characteristics

Panel A: Correlations Between $Turn_{it+k}$ and $ Ret_{it+k} $ on Day $t+k$ Sorted by Type of News on Day t						
Characteristic	Rank	Day $t-10$ Correlation	Day t Correlation	Day $t+10$ Correlation	Post-Event Difference	Pre-Post Event Difference
News on day t	All	0.298 (0.002)	0.441 (0.002)	0.288 (0.002)	-0.153** (0.002)	-0.010** (0.002)
No news on day t	All	0.269 (0.001)	0.239 (0.001)	0.271 (0.001)	0.032** (0.001)	0.002** (0.001)
	Diff				-0.185** (0.002)	-0.012** (0.002)
Earnings news on day t	All	0.298 (0.004)	0.583 (0.003)	0.295 (0.011)	-0.290** (0.005)	-0.004 (0.005)
Nonearnings news on t	All	0.299 (0.002)	0.395 (0.002)	0.288 (0.002)	-0.107** (0.002)	-0.011** (0.002)
	Diff				-0.201** (0.005)	0.009 (0.006)
$MsgSA_{it}$ (News on day t)	High	0.335 (0.004)	0.616 (0.003)	0.302 (0.004)	-0.315** (0.004)	-0.034** (0.004)
	Low	0.291 (0.002)	0.359 (0.002)	0.284 (0.002)	-0.075** (0.003)	-0.007** (0.002)
	Diff				-0.302** (0.005)	-0.025** (0.005)
Panel B: Average Price Impact on Day $t+k$ Sorted by Firm Characteristic on Day t						
Characteristic	Rank	Day $t-10$ Price Impact	Day t Price Impact	Day $t+10$ Price Impact	Post-Event Difference	Pre-Post Event Difference
$Size_{it}$ (News on day t)	High	1.83 (0.02)	1.78 (0.02)	1.83 (0.02)	0.05** (0.01)	0.00 (0.01)
	Low	1.69 (0.02)	1.55 (0.02)	1.64 (0.02)	0.09** (0.02)	-0.04* (0.02)
	Diff				-0.04* (0.02)	0.04* (0.02)
$AnalystsA_{it}$ (News on day t)	High	1.64 (0.02)	1.58 (0.02)	1.63 (0.02)	0.05** (0.01)	-0.01 (0.01)
	Low	1.71 (0.01)	1.66 (0.01)	1.71 (0.02)	0.05** (0.02)	0.00 (0.02)
	Diff				0.00 (0.02)	0.00 (0.02)
$DispersA_{it}$ (News on day t)	High	1.46 (0.02)	1.42 (0.01)	1.45 (0.01)	0.04** (0.01)	-0.01 (0.01)
	Low	1.86 (0.02)	1.76 (0.01)	1.86 (0.02)	0.10** (0.01)	0.01 (0.01)
	Diff				-0.05** (0.01)	-0.01 (0.01)

(continued)

Table 6
Continued

$InstOwnSA_{it}$ (News on day t)	High	1.19 (0.01)	1.10 (0.01)	1.15 (0.01)	0.05** (0.01)	-0.03* (0.01)
	Low	2.53 (0.02)	2.45 (0.02)	2.54 (0.02)	0.09** (0.02)	0.01 (0.04)
	Diff				-0.03 (0.02)	-0.03 (0.02)
$IlliquiditySA_{it}$ (News on day t)	High	2.65 (0.02)	2.78 (0.02)	2.58 (0.02)	-0.20** (0.03)	-0.07** (0.03)
	Low	1.50 (0.01)	1.36 (0.01)	1.50 (0.01)	0.14** (0.01)	0.00 (0.01)
	Diff				-0.34** (0.03)	-0.09** (0.03)
$IlliquiditySA_{it}$ (No news on t)	High	3.11 (0.02)	3.44 (0.02)	3.03 (0.02)	-0.40** (0.01)	-0.07** (0.01)
	Low	1.04 (0.01)	0.91 (0.01)	1.06 (0.01)	0.15** (0.01)	0.03** (0.01)
	Diff				-0.54** (0.01)	-0.10** (0.01)
$IlliquiditySA_{it}$ (News vs. none)	DDiff				0.20** (0.04)	0.01 (0.04)

Table 6 reports only the correlation coefficients based on Equation (17) in the text:

$$Turn_{it} = Intercept + Slope * |Ret_{it}| + e_{it} \text{ for all } i \text{ on each day } t.$$

The dependent variable is abnormal daily turnover ($Turn_{it}$); and the independent variable is absolute daily returns $|Ret_{it}|$. The first three columns indicate the correlation between absolute returns and volume ten days prior to news ($t-10$), on the day of news (t), and ten days after news ($t+10$). I examine *changes* in the correlation between absolute returns and volume before and after the same news events in the last two columns. I obtain these estimates from regressions of Equation (17) in three different subsamples that include absolute returns and volume for firms with news events exactly k days ago, where k is equal to either -10, 0, or +10 days relative to the news event. The post-event difference is the correlation on day $t+10$ minus the correlation on day t , whereas the pre-post event difference is the correlation on day $t+10$ minus the correlation on day $t-10$. I estimate Equation (17) using daily cross-sectional regressions. In parentheses, I report Newey and West (1987) standard errors that allow for autocorrelation of up to three daily lags in the time series of coefficients. The rows in the panels indicate the news or firm characteristic subsample in which I estimate the correlation between absolute returns and volume. Panel A compares news days versus nonnews days, earnings news days versus nonearnings news days, and news days with news messages in the top vs. bottom quintile. Panel B compares the top and bottom quintiles of size, analyst coverage, analyst forecast dispersion, institutional ownership, and illiquidity. The last seven rows analyze correlations around no-news events in both illiquidity quintiles and the difference-in-difference (DDiff) in correlations between the two liquidity quintiles for news and nonnews events. The * and ** symbols denote statistical significance at the 5% and 1% levels. I suppress all asterisks for the raw correlations.

- The news-day increase in the return–volume correlation is much larger for high-message news than for low-message news.
- The event-time drop after the news is especially strong among small and illiquid firms.
- The evidence is much less supportive of a pure attention or disagreement-based explanation.

Economic message. The strongest comovement appears when news seems most informative, not merely most attention grabbing.

5.7 Table 7 and Figure 4: Price impact around news

Table 7
Intraday price impact sorted by firm and news characteristics

Panel A: Average Price Impact on Day $t + k$ Sorted by News Characteristic on Day t						
Characteristic	Rank	Day $t - 10$ Price Impact	Day t Price Impact	Day $t + 10$ Price Impact	Post-Event Difference	Pre-Post Event Difference
News on day t	All	1.66 (0.01)	1.58 (0.01)	1.64 (0.01)	0.06** (0.01)	-0.01 (0.01)
No news on day t	All	1.77 (0.01)	1.78 (0.01)	1.77 (0.01)	-0.01** (0.00)	0.00 (0.00)
	Diff				0.07** (0.01)	-0.01 (0.01)
Earnings news on day t	All	1.74 (0.02)	1.56 (0.01)	1.69 (0.02)	0.13** (0.01)	-0.05** (0.02)
Nonearnings news on t	All	1.64 (0.01)	1.59 (0.01)	1.63 (0.01)	0.04** (0.02)	-0.01 (0.02)
	Diff				0.08** (0.02)	-0.05** (0.02)
MsgSA $_{it}$ (News on day t)	High	1.85 (0.02)	1.62 (0.02)	1.79 (0.02)	0.17** (0.01)	-0.06** (0.01)
	Low	1.60 (0.01)	1.56 (0.01)	1.59 (0.01)	0.03** (0.01)	-0.01 (0.01)
	Diff				0.13** (0.01)	-0.05** (0.01)
Panel B: Average Price Impact on Day $t + k$ Sorted by Firm Characteristic on Day t						
Characteristic	Rank	Day $t - 10$ Price Impact	Day t Price Impact	Day $t + 10$ Price Impact	Post-Event Difference	Pre-Post Event Difference
Size $_{it}$ (News on day t)	High	1.83 (0.02)	1.78 (0.02)	1.83 (0.02)	0.05** (0.01)	0.00 (0.01)
	Low	1.69 (0.02)	1.55 (0.02)	1.64 (0.02)	0.09** (0.02)	-0.04* (0.02)
	Diff				-0.04* (0.02)	0.04* (0.02)
AnalystsSA $_{it}$ (News on day t)	High	1.64 (0.02)	1.58 (0.02)	1.63 (0.02)	0.05** (0.01)	-0.01 (0.01)
	Low	1.71 (0.01)	1.66 (0.01)	1.71 (0.02)	0.05** (0.02)	0.00 (0.02)
	Diff				0.00 (0.02)	0.00 (0.02)
DispersSA $_{it}$ (News on day t)	High	1.46 (0.02)	1.42 (0.01)	1.45 (0.01)	0.04** (0.01)	-0.01 (0.01)
	Low	1.86 (0.02)	1.76 (0.01)	1.86 (0.02)	0.10** (0.01)	0.01 (0.01)
	Diff				-0.05** (0.01)	-0.01 (0.01)

Table 7
Continued

InstOwnSA _{it} (News on day t)	High	1.19 (0.01)	1.10 (0.01)	1.15 (0.01)	0.05** (0.01)	-0.03* (0.01)
	Low	2.53 (0.02)	2.45 (0.02)	2.54 (0.02)	0.09** (0.02)	0.01 (0.04)
	Diff				-0.03 (0.02)	-0.03 (0.02)
IlliquiditySA _{it} (News on day t)	High	2.65 (0.02)	2.78 (0.02)	2.58 (0.02)	-0.20** (0.03)	-0.07** (0.03)
	Low	1.50 (0.01)	1.36 (0.01)	1.50 (0.01)	0.14** (0.01)	0.00 (0.01)
	Diff				-0.34** (0.03)	-0.09** (0.03)
IlliquiditySA _{it} (No news on t)	High	3.11 (0.02)	3.44 (0.02)	3.03 (0.02)	-0.40** (0.01)	-0.07** (0.01)
	Low	1.04 (0.01)	0.91 (0.01)	1.06 (0.01)	0.15** (0.01)	0.03** (0.01)
	Diff				-0.54** (0.01)	-0.10** (0.01)
IlliquiditySA _{it} (News vs. none)	DDiff				0.20** (0.04)	0.01 (0.04)

For firms and news events with different characteristics, Table 7 reports firms' average share-weighted price impact before, during, and after news and non-news events. Share-weighted price impact is based on total signed order flows and returns measured at 5-minute intraday intervals. I use the Lee and Ready (1991) algorithm to sign total order flow in the ISSM and TAQ databases. Price impact is the ratio of the covariance of share-weighted signed order flow and returns to the variance of share-weighted signed order flows. The share-weighted price impact measure is equal to the percentage price impact of transacting one percent of the firm's total shares outstanding. The first three columns indicate the average price impact of order flow ten days prior to a firm event ($t - 10$), on the event day (t), and ten days after the event ($t + 10$). I examine *changes* in price impact before and after the events in the last two columns. In parentheses, I report Newey and West (1987) standard errors that allow for autocorrelation of up to three daily lags in the time series of average price impacts. The rows in the two panels indicate the news or firm characteristic subsample in which I estimate average price impact around news or nonnews events. Panel A compares price impact on news days versus nonnews days, earnings news days versus nonearnings news days, and news days with newswire messages in the top vs. bottom quintile. Panel B compares price impact around news in the top and bottom quintiles of size, analyst coverage, analyst forecast dispersion, institutional ownership, and illiquidity. The last seven rows analyze price impact around no-news events in both illiquidity quintiles and the difference-in-difference (DDiff) in correlations between the two liquidity quintiles for news and nonnews events. The * and ** symbols denote statistical significance at the 5% and 1% levels. I suppress all asterisks for the raw price impacts.

Read:

- Price impact falls on the news day relative to surrounding days.
- Figure 4 shows a sharp dip centered on the event date and a rebound afterward.
- The postnews increase in price impact is larger for high-message news, illiquid firms, and small firms.

Economic message. Once public news resolves some information asymmetry, the same signed order flow moves prices less.

6 Short summary

Tetlock studies whether public financial news resolves asymmetric information. He builds a three-period model with informed traders who possess a private signal and face a persistent liquidity

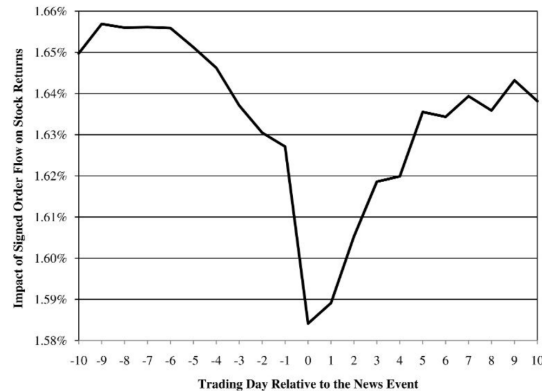


Figure 4

The price impact of order flow before and after public news events

For each of the ten days before and after firm news events, Figure 4 reports the daily price impact of signed order flow. I use the Lee and Ready (1991) algorithm to sign order flows as buyer initiated or seller initiated. In each five-minute interval in a trading day, I aggregate share-weighted order flows – as a percentage of shares outstanding – and calculate stock returns. Price impact is the covariance of five-minute returns and five-minute order flows, divided by the variance of five-minute order flows. This figure shows the cross-sectional average of price impact for all firms experiencing a news event k days ago, where k is an integer ranging from -10 to $+10$. See text for further estimation details.

shock. Public news reveals the signal, allowing previously uninformed traders to infer more about the shock and absorb it more aggressively.

The model predicts four facts: less reversal after news, more continuation on high-volume news days, stronger contemporaneous comovement between absolute returns and turnover, and lower price impact of signed order flow around news. Using Dow Jones and *WSJ* firm-level news matched to CRSP, Compustat, I/B/E/S, 13F, and intraday trade data, the paper finds strong support for all four predictions.

The main conceptual contribution is that public news does not merely move prices; it changes the market’s information structure. The main empirical contribution is to show that return predictability, volume, and price impact all line up with this mechanism.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that public financial news tends to *resolve asymmetric information*. It does so in a way that reduces short-run reversals, raises news-day continuation when turnover is high, strengthens the relation between absolute returns and trading volume, and lowers the price impact of order flow.

7.2 Economic intuition

Before the news arrives, informed traders know more than others and trade partly on that informational advantage. When the news is released, uninformed traders learn enough to infer more about the motives behind prior order flow. This reduces the informational rent in informed trading, makes liquidity provision more aggressive, and causes the outstanding liquidity shock to be absorbed gradually rather than instantly.

7.3 Takeaway

The cleanest way to remember the paper is: **public news helps the market figure out what informed traders were already partly reacting to**. Once that happens, prices continue in the news direction, volume and returns become more tightly linked, and signed order flow becomes less powerful at moving prices. That is a coherent asymmetric-information story, and the paper provides unusually broad evidence for it.